Pitch Vernier (battery movement)

The Slocum doesn't have elevators or a moving tail to set its dive angle — it **slides its heaviest internal mass back and forth.** A **lead screw drives the forward ~10 kg battery pack fore or aft**, shifting the centre of gravity to trim the vehicle's pitch. The manual calls this the **Pitch Vernier**: the ballast pump supplies the *buoyancy* that makes the glider dive or climb, and the moving battery is the **fine adjustment** that sets *at what angle* it does so.

- Moving the battery **forward** makes the nose **heavy** → **the glider pitches down (dive)**.
- Moving the battery **aft** lifts the nose → **the glider pitches up (climb)**.
- On the surface the battery is driven **all the way forward** to raise the tail (and its antennas) out of the water for comms.

The vehicle is designed to dive and climb at about **26°**. This only works if the glider is properly trimmed — the **H-moment must be 6 mm ±1** (see *H Moment Calculation / Adjusting the H Moment* in the Maintenance Manual) — otherwise the ballast pump's moment won't pitch the glider as expected and the battery vernier has to fight it.

Source

Paraphrased from the *Slocum G3 Glider Operators Manual* (Rev. 1, "Pitch Vernier" and the sample mission appendix), the *Slocum G3 Maintenance Manual* (forward-section assembly), the UG2 community Slack, and the Teledyne Webb Research user forum. Condensed field reference — always defer to the official Teledyne documentation, and contact `glidersupport@teledyne.com` before changing pitch or trim configuration.